Chapter 1

Introduction

Text before the first section. With some inline mathematics $f(x) = \int_0^x e^{-t^2} dt$. That looks nice.

And some displayed mathematics:

$$f(x) = \int_0^x e^{-t^2} dt.$$

Chapter 2

Sequences

Motivation and context remarks.

2.1 Monotonic and bounded sequences

Before considering the *convergence* of infinite sequences, we consider the possible increasing, decreasing and bounded behaviour of sequences.

Definition 2.1.1 An increasing sequence $\{x_n\}$ is one whose elements satisfy

$$x_1 \le x_2 \le x_3 \le \dots.$$

A decreasing sequence $\{y_n\}$ is one whose elements satisfy

$$y_1 \ge y_2 \ge y_3 \ge \dots.$$

If the inequalities are strict inequalities then we would say the sequence is strictly increasing, or strictly decreasing, as appropriate. Or the inequalities might only hold true from some element x_k onwards, in which case we would say the sequence is increasing/decreasing *from* k, as appropriate.

The term *monotonic*, refers to a sequence that is either increasing or decreasing. Similarly for *strictly monotonic*. ◊

If one has a nice formula for a sequence then the quickest way to see of the sequence is monotonic might be to plot some elements of the sequence using a computer. There are many ways to do this. Here is a quick example of how to do this using Python and the Matplotlib library.

```
import matplotlib.pyplot as plt
S = [(n**2 + 2)/2**n for n in range(1,50)]
plt.plot(range(1,50),S,'o')

# Add some labels and a title for clarity
plt.xlabel("Index")
plt.ylabel("Sequence element")

# Display the plot
plt.show()
```

Here is an alternative way, and somewhat quicker, using the Sage language.

```
S = [(n^2 + 2)/2^n for n in range(1,50)]
list_plot(S)
```

Investigating the first few elements of a sequence like this may provide evidence that a sequence is monotonic. However a more rigorous approach, that addresses *all* elements of the sequence, is needed if we want to formally prove that the sequence is monotonic. There are, at least, two approaches we can take to this. Suppose we are given a sequence $\{a_n\}_{n=1}^{\infty}$ and we wish to investigate its monotonic nature. By considering the difference or ratio of consecutive terms, and comparing this to 0 or 1, we can draw conclusions about the monotonic nature of the sequence.

1. Consider the general difference, $a_{n+1} - a_n$, between consecutive terms, and then compare this to 0, noting that

   $$\{a_n\} \text{ is decreasing if and only if for all } n \geq 1 \text{ we have } a_{n+1} - a_n \leq 0,$$

   and

   $$\{a_n\} \text{ is increasing if and only if for all } n \geq 1 \text{ we have } a_{n+1} - a_n \geq 0.$$

   Of course, strictly increasing/decreasing versions of these results hold if the inequalities are strict.

2. Provided the sequence elements are positive and never zero, consider the general ratio, a_{n+1}/a_n, between consecutive terms, and then compare this to 1, noting that

   $$\{a_n\} \text{ is decreasing if and only if for all } n \geq 1 \text{ we have } \frac{a_{n+1}}{a_n} \leq 1,$$

   S = [(n^2 + 2)/2^n for n in range(1,50)] list_plot(S) and

   $$\{a_n\} \text{ is increasing if and only if for all } n \geq 1 \text{ we have } \frac{a_{n+1}}{a_n} \geq 1.$$

   Of course, strictly increasing/decreasing versions of these results hold if the inequalities are strict.

Which approach to use might be based on personal preference or the particular form of the formula defining the sequence element.

Example 2.1.2 An increasing, then decreasing sequence. Investigate the sequence $\{a_m\}_{m=1}^{\infty}$ defined by

$$a_m = \frac{5^m}{m!},$$

and describe its increasing/decreasing behaviour.

Solution. Consider the ratio of consecutive elements.

$$\begin{aligned} \frac{a_{m+1}}{a_m} &= \frac{5^{m+1}}{(m+1)!} \frac{n!}{5^m} \\ &= \frac{5}{m+1}. \end{aligned}$$

The quantity $5/(n+1)$ can be compared to 1. It is greater than 1 for $n < 4$ and less than 1 for $n \geq 5$. So the sequence increases at first, but then decreases for all $n \geq 5$.

```
S = [5^n / factorial(n) for n in range(0,20)]
list_plot(S)
```

□